Project 2 Design Report

Our project uses DV routing to allow nodes to community across many hops. We implemented the DV algorithm in DistanceVectorP.nc and IP forwarding in IPP.nc . DV uses neighbor discovery interface to find directly connected routes and the IP module uses Routing.getRoute() to find the next hop when forwarding the packet. Each node has a routing table that has: a destination, newt hop, hop count, validity flag, and routing age. Cost of zero is its route to itself, one is the cost to its directly connected neighbor and 16 means it is unreachable. The routing timer runs every three seconds and each update the node refreshes its direct neighbors, checks for failed neighbors, ages learned routes and sends its DV table to its neighbors. Learned routes expire after four routing periods without being refreshed, it then gets marked with a cost of 16. We implemented poison reverse to reduce routing loops. When advertising to a neighbor that is currently being used as the next hop for a destination, the node advertises that destination with a cost of 16. Other neighbors receive the normal route cost. Each DV advertisement uses a 2-byte header and 3 bytes per routing entry. When sending or forwarding a packet, the IP module checks the routing table for the destination and forwards the packet to the selected next hop. The forwarding process decreases the and ping replies are routed back to the original source using the same routing mechanism.